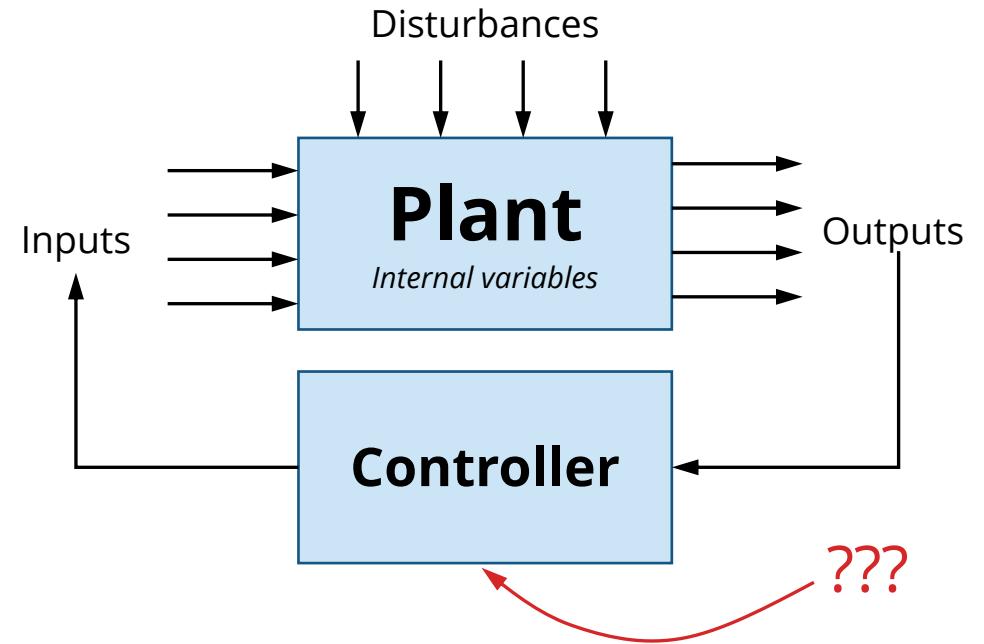




AA 513 Multivariable Control

Lecture 05 // Week 5

Instructor: Prof. Karen Leung

Announcements

- Exercises 9, 10, 11, 12 out (soon)
- Self-assessments due this week
- Check-in sessions by appointment to discuss research papers and tutorial topics.
- Research paper presentation
 - Follow PACES framework
 - Shared slide deck released
- Tutorial submission
 - **Submission:** Upload video to shared folder
 - **Peer review:** You will be assigned videos to review. You are welcome to review more if you like.

Week 10 schedule

Schedule

- 6 minutes per talk + 2 minute questions

Session 1			Session 2			Session 3		
#	Time	Name	#	Time	Name	#	Time	Name
1	6:00PM		8	7:00PM		15	8:00PM	
2	6:08PM		9	7:08PM		16	8:08PM	
3	6:16PM		10	7:16PM		17	8:16PM	
4	6:24PM		11	7:24PM		18	8:24PM	
5	6:30PM		12	7:30PM		19	8:30PM	
6	6:38PM		13	7:38PM		20	8:38PM	
7	6:46PM		14	7:46PM		21	8:46PM	
						22	8:54PM	

Last time

- Guest lecture by Dr Liam Krause, topic on Markov Decision Process and algorithms for finding optimal policies
 - **MDP:** More general framework for modeling systems that evolve over time
 - Assumes transition *probabilities*
 - Expectation over transition probabilities
 - **Value iteration:** DP until convergence
 - **Policy iteration:** Iterate between policy evaluation & policy improvement
 - **POMDPs:** *Partially observable* MDPs. Keep a *belief* over your state.
 - Expectation over transition AND belief probabilities
 - **Street-fighting techniques:** Value function approximations, MCTS (local search)

State Space $\mathcal{S}$

Action Space $\mathcal{A}$

Transition Model $T(s' | s, a)$

Reward Function $R(s, a)$

This week

- Continuous time: Hamilton-Jacobi-Bellman equation
- Linear quadratic regulator
- No lecture next week (Veteran's Day)
 - Will release recording

Exercises

- Ex9 Stochastic Bellman equation
- Ex10: Finite horizon LQR
- Ex11: Infinite horizon LQR
- Ex12: Tracking LQR

Continuous-time, continuous state & control

- Consider cost $J_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$
- Cast as discrete time problem with timestep ϵ
 - $J_{discrete}(\mathbf{x}_t, \mathbf{u}_t, t) = \int_t^{t+\epsilon} g(\mathbf{x}(\tau), \mathbf{u}(\tau), \tau) d\tau \approx \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t)$, for small ϵ

$$V^*(\mathbf{x}(t), t) = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + V^*(\mathbf{x}(t + \epsilon), t + \epsilon)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + V^*(\mathbf{x}(t + \epsilon), t + \epsilon) - V^*(\mathbf{x}(t), t)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + (V^*(\mathbf{x}(t), t) + \nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2) - V^*(\mathbf{x}(t), t)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \frac{1}{\epsilon} (\nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2))$$

$$0 = \lim_{\epsilon \rightarrow 0} \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) + \frac{\partial V}{\partial t} + O(\epsilon)$$

$$0 = \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)$$

Hamilton-Jacobi-Bellman equation

Consider cost $g_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \{g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)\} = 0$$

Terminal condition: $V(\mathbf{x}, T) = g_N(\mathbf{x})$

Solve PDE backward in time. But generally not easy to solve ☹

What have we just done?

- Derived the key equations describing the optimal solution to a sequential decision-making problem
 - Under some assumptions, these are necessary and sufficient conditions for optimality
- Solving exactly is hard / intractable
- Instead, perform “**approximate** dynamic programming” and **function approximators** for value function

Can we ever hope to solve these equations exactly?!

- Yes! Under strict assumptions on the functional form of dynamics, cost, and value function.
- This is known as the linear quadratic regulator!

The discrete-time finite horizon LQR problem

Regulate to zero, want state and control to be small

$$\min_{\mathbf{u}_{0:T-1}} \quad \mathbf{x}_T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t Q_t \mathbf{x}_t + \mathbf{u}_t R_t \mathbf{u}_t \quad \text{Quadratic cost} \quad (1)$$

subject to $\mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \quad (2)$

Linear dynamics

$$\mathbf{x}_0 = \mathbf{x}_{\text{current}} \quad (3)$$

Initial state

* Not really a constraint since state is not an optimization variable. More to just “anchor” the problem to a starting state.

Some notes about the LQR problem

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T^T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t^T Q_t \mathbf{x}_t + \mathbf{u}_t^T R_t \mathbf{u}_t \\ \text{subject to} \quad & \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

- No constraints! (aside from dynamics)
- Cost and dynamics can be ***time-varying***
- Is it always possible to regulate to zero? (i.e., stabilize system?)
 - Can we even control it?
 - What if there are uncontrollable/unobservable modes?
- Conditions on the cost matrices.

How to solve the LQR problem?

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T^T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t^T Q_t \mathbf{x}_t + \mathbf{u}_t^T R_t \mathbf{u}_t \\ \text{subject to} \quad & \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

- Is the problem convex?
- Can we solve it in closed-form?

Recall: Bellman equation

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u} \in \mathcal{U}(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t + 1), \quad V^*(\mathbf{x}_T, T) = g_T(\mathbf{x}_T)$$

$$g_T(\mathbf{x}_T)$$

$$f(\mathbf{x}_t, \mathbf{u}_t, t)$$

$$g(\mathbf{x}_t, \mathbf{u}_t, t)$$

$$\mathbf{u} \in \mathcal{U}(\mathbf{x}_t)$$

Simplifying assumptions: Not time-varying

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Discrete-time finite horizon LQR (time-invariant)

1. set $P_T = Q_T$

2. for $t = N, \dots, 1$,

$$P_{t-1} = Q + A^T P_t A - A^T P_t B (R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \dots, N - 1$, $\mathbf{u}_t^* = K_t \mathbf{x}_t$

Discrete-time finite horizon LQR (time-varying)

What changes?

1. set $P_T = Q_T$
2. for $t = N, \dots, 1$,

$$P_{t-1} = Q + A^T P_t A - A^T P_t B (R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \dots, N - 1$, $\mathbf{u}_t^* = K_t \mathbf{x}_t$

Discrete-time **infinite** horizon LQR (time-invariant)

What changes?

1. set $P_T = Q_T$
2. for $t = N, \dots, 1$,

$$P_{t-1} = Q + A^T P_t A - A^T P_t B (R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \dots, N - 1$, $\mathbf{u}_t^* = K_t \mathbf{x}_t$

What about the continuous-time setting?

- Same process but with the HJB equation!

$$0 = \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in \mathcal{U}(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)$$

$$\min_{\mathbf{u}(\cdot)} \quad \mathbf{x}(T)Q_T\mathbf{x}(T) + \int_0^T \mathbf{x}(t)Q(t)\mathbf{x}(t) + \mathbf{u}(t)R(t)\mathbf{u}(t) \quad (1)$$

$$\text{subject to} \quad \dot{\mathbf{x}} = A(t)\mathbf{x}(t) + B(t)\mathbf{u}(t), \quad t = [0, T] \quad (2)$$

$$\mathbf{x}(0) = \mathbf{x}_{\text{current}} \quad (3)$$

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Continuous-time finite horizon LQR (time-invariant)

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time finite horizon LQR (time-varying)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time **infinite** horizon LQR (time-invariant)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Other variations

- Regulating to $(\mathbf{x} - \mathbf{x}_g)^T Q (\mathbf{x} - \mathbf{x}_g) + (\mathbf{u} - \mathbf{u}_g)^T R (\mathbf{u} - \mathbf{u}_g)$
 - Cost has linear terms
 - Value function takes on the form: quadratic + linear + constant
 - Recursion updates for all terms
 - Controller is affine
- Tracking LQR $(\mathbf{x}_t - \tilde{\mathbf{x}}_t)^T Q (\mathbf{x}_t - \tilde{\mathbf{x}}_t) + (\mathbf{u}_t - \tilde{\mathbf{u}}_t)^T R (\mathbf{u}_t - \tilde{\mathbf{u}}_t)$
 - Have a desired trajectory to follow
- Iterative LQR
 - Iteratively approximate non-LQR problem as LQR problem and solve several times

Tracking LQR

- Consider desired state and control sequence
 - $(\tilde{\mathbf{x}}_0, \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_{T-1}), (\tilde{\mathbf{u}}_0, \tilde{\mathbf{u}}_1, \dots, \tilde{\mathbf{u}}_{T-1})$
- Consider *error* states, controls, and dynamics
 - $\delta \mathbf{x}_t = \mathbf{x}_t - \tilde{\mathbf{x}}_t$
 - $\delta \mathbf{u}_t = \mathbf{u}_t - \tilde{\mathbf{u}}_t$
 - $\delta \dot{\mathbf{x}}_t = \dot{\mathbf{x}}_t - \dot{\tilde{\mathbf{x}}}_t = A_t \mathbf{x}_t + B_t \mathbf{u}_t - (A_t \tilde{\mathbf{x}}_t + B_t \tilde{\mathbf{u}}_t) = A_t \delta \mathbf{x}_t + B_t \delta \mathbf{u}_t$
- Want error $\rightarrow 0$
 - Apply standard LQR on *error* states, controls, and dynamics
 - $\delta \mathbf{u}_t^* = K_t \delta \mathbf{x}_t \Rightarrow \mathbf{u}_t^* = K_t \delta \mathbf{x}_t + \tilde{\mathbf{u}}_t$